

Privacy-preserving federated learning for early autism spectrum disorder detection using MobileNet and temporal convolutional networks

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ABSTRACT

Early autism spectrum disorder (ASD) diagnosis is essential to enhance developmental outcomes, but the conventional methods of diagnosis are subjective, resource-consuming, and delayed. Deep learning has demonstrated good opportunities in automated screening of ASD; however centralized training is extremely privacy concerns on critical pediatric data. The proposed paper presents a federated learning system, which combines MobileNet that handles lightweight facial feature extraction with a Temporal Convolutional Network (TCN) that can handle temporal patterns modeling. A federated averaging protocol allows collaborative learning among distributed clients without having to share raw data, and differential privacy (DP) further supports this. Testing of an ASD face dataset reveals that the framework has 87.50% accuracy with IID distribution with a fairness score of 0.875, and 74.00 percent accuracy with DP-enabled training, with fairness to privacy-utility trade-offs. Findings emphasize the practicality of scalable ASD detection systems that are privacy-preserving and demonstrates potential for deployment in real-world clinical settings, subject to further large-scale validation.

Introduction

Autism Spectrum Disorder (ASD) is an intricate developmental disorder of the nervous system, occurring in about 1 out of every 36 children globally, where there are indefinite impairments in social interaction, communication and limited repetitive behaviors¹. Early intervention and diagnosis have a high positive impact on the long-term developmental results, and the best intervention is before the age of 3². Nevertheless, the current diagnosis methods are significantly limited such as the lack of objective assessment criteria, prolonged examination procedures, and the absence of appropriate access to expert clinicians.

Conventional ASD screening involves extensive use of standardized tests like the Modified Checklist for Autism in Toddlers-Revised with Follow-up (M-CHAT-R/F)³, Social Communication Questionnaire (SCQ) and full-scale clinical interviews. Although clinically proven, these procedures are subjective in nature, culturally biased, and also have vast delays between first concern and final diagnosis, which in most instances take more than 18 months⁴.

The development of artificial intelligence and deep learning has proposed promising alternatives of automated ASD detection. Recent research shows that convolutional neural networks (CNNs) can detect fine craniofacial features and behavioral patterns that predict autism by analyzing images and videos^{5,6}. Nonetheless, implementation of such systems in the healthcare sector is faced with significant obstacles pertaining to privacy of data, compliance with regulations and institutional fragmentation of medical information.

Federated Learning (FL) is a paradigm shift, which has the potential to solve these obstacles by facilitating joint model training that does not centralize sensitive patient information⁷. This distributed learning model enables numerous healthcare organizations to collaborate with the development of models and retain a high level of data locality and data privacy.

Traditional ASD Diagnostic Approaches

History of ASD diagnostic procedures has undergone a series of screening devices and a number of generations of clinical treatment procedures. Such early diagnosis instruments like the Checklist for Autism in Toddlers (CHAT) and its later developments like M-CHAT and M-CHAT-R/F have already set the standardized procedures of screening at the population level. These tests have medium sensitivity (0.85-0.95) and specificity (0.84-0.99) in a controlled clinical study.

Detailed diagnostic evaluations, such as the Autism Diagnostic Observation Schedule (ADOS-2) and Autism Diagnostic Interview-Revised (ADI-R) are the current gold standard of ASD diagnosis. Yet, such methods demand a high level of clinical

experience, consume a great deal of time (2-4 hours assessment time), and exhibit high inter-rater reliability even in the hands of well-trained clinicians⁸.

Recent meta-analyses underscore recurrent issues with conventional methods of diagnosis, such as cultural bias in assessments tools, lack of access to underserved groups, and high rates of diagnostic latency of 14-30 months between the onset of parents with parental concerns^{9,10}.

Deep Learning Approaches in ASD Detection

Neuroimaging-Based Approaches

Convolutional neural networks used on structural MRI data have been able to classify between ASD and neurotypical development with over 90% accuracy¹¹. Heinsfeld et al.¹² showed that deep learning models were capable of detecting ASD-related patterns in the resting functional connectivity with 70 percent accuracy on various datasets. Nevertheless, neuroimaging has some drawbacks such as its high cost, lack of accessibility among pediatrics, and the need to access special equipment.

Facial Analysis Methods

More recent studies have also found some slight craniofacial morphological differences that are linked to ASD, as a result there has been the establishment of facial recognition systems that can be used to screen people automatically. The VGG-16 networks by Tao et al.¹³ were 94.5% accurate on standardized face photographs. Alkahtani et al.¹⁴ showed that MobileNet architectures were able to detect ASD-related facial features with 92.3-percent accuracy and was computationally efficient enough to use on a mobile system.

Behavioral Pattern Recognition

Video-based behavioral analysis is an area of promising advancement of detecting ASD. Zunino et al.¹⁵ came up with CNN-LSTM hybrid models that reached 85 percent accuracy in detecting repetitive behaviors and social attention patterns. Temporal Convolutional Networks have demonstrated some promising results with regard to capturing long range temporal aspects in sequences of behavioral activity¹⁶.

Multimodal Integration

Recent methods use several modalities of data to enhance diagnostic accuracy. Eslami et al.¹⁷ revealed that a combination of facial, vocal and behavioral features was found to be 96.8% accurate in classifying ASD compared to single modality methods.

Privacy-Preserving Healthcare AI

The implementation of AI systems in the healthcare industry must have effective privacy-saving technologies to safeguard the confidential details of patients and enable cooperative research and model building. The former methods, including data de-identification, anonymization and computationally secure multi-party approaches, have served as a basis in protecting privacy but are becoming ineffective, against new advances in re-identification¹⁸. Healthcare big data breaches only underscore the weaknesses of these traditional approaches¹⁹. Introduced by Dwork²⁰, differential privacy provides more powerful guarantees by introducing noise to query responses or model coefficients, which are calibrated. Recent applications in the healthcare AI have demonstrated encouraging outcomes, which allow differentially private machine learning with a practical utility-privacy trade-off^{21,22}.

Simultaneously, federated learning has been on the rise as one of the potential solutions to the problem of privacy and data fragmentation in healthcare. FL ensures that data can be kept locally and only model updates are shared, therefore, it enables multi-institutional training without breaking patient confidentiality. Li et al.²³ showed that federated training was successful in medical image classification in hospitals with a minimum performance degradation. The recent surveys have also pointed out its advantages in pediatric healthcare, where regulatory limitations on information sharing are the strictest, which is why federated strategies can be of the greatest value²⁴.

Federated Learning Architecture

The FedAvg algorithm presented by McMahan et al.²⁵ allows training in a distributed fashion by using local updates and global aggregation. Follow-up systems such as FedProx²⁶, FedNova²⁷ and SCAFFOLD²⁸ deal with the issue of non-IID data distribution and client heterogeneity.

Generalized FedAvg aggregation with client weights: Measured as:

$$w_{t+1} = \sum_{k=1}^K \frac{n_k}{\sum_{j=1}^K n_j} w_t^{(k)} \quad (1)$$

Federated learning services in healthcare have overcome domain-specific considerations such as institutional and regulatory data governance and clinical validation needs. In healthcare AI, federated methods have been applied very widely, including oncology²⁹, radiology³⁰ and precision medicine³¹.

More-enhanced privacy preserving methods such as secure aggregation, homomorphic encryption, and integration of differential privacy offers further protection of gradient-based inference attacks and model inversion vulnerability.

Research Gaps and Motivation

Despite significant advances in both ASD detection technologies and privacy-preserving machine learning, several critical gaps remain:

Limited Federated ASD Detection

Existing ASD detection systems primarily rely on centralized training, limiting their applicability in privacy-sensitive healthcare environments.

Computational Efficiency

Current deep learning approaches often require substantial computational resources, limiting deployment in resource-constrained clinical settings.

Temporal Pattern Modelling

While CNNs excel with non-IID data distribution and communication constraints, They fail to capture temporal behavior pattern relevant to ASD.

Fairness and Equity

Limited analysis of model performance equity across diverse client populations and institutional settings.

Responsible Regulations and Ethics

The structure of the framework is grounded on the regulation of the functioning of medical AI according to the rules of the sphere of health care and its ethical issues. The privacy-preserving federation model eliminates significant concern on data privacy of patients whilst the interpretable attention processes provide transparency in areas that transparency is required in clinical decision support systems⁴. It is the message of medical ethics that prioritizes safety of the patient and minimizes potential harm it may cause due to missed diagnosis that has been identified as a reason why medical practitioners approach class imbalance and high sensitivity conservatively.

We have made our contribution in the form of a new Federated MobileNet-TCN architecture aimed at ASD detection with privacy guarantee. A combination of MobileNet lightweight architecture and Temporal Convolutional Networks (TCNs) is designed to meet the task of both machine efficiency demanded by edge deployment and the complexity of recognition of temporal patterns demanded during behavioral analysis. Moreover, we introduce the use of differential privacy controls to give mathematical guarantees against privacy attacks.

The following is a summary of this work's primary contributions:

- For ASD detection in distributed healthcare settings, we suggest a novel Federated MobileNet–TCN architecture that combines efficient temporal modeling with lightweight spatial feature extraction.
- In order to offer formal privacy guarantees against data leakage and facilitate cooperative model training, we incorporate differential privacy into the federated learning pipeline.
- In order to replicate realistic multi-institutional healthcare settings, we assess the suggested framework under both IID and non-IID data distributions.
- We conduct a thorough analysis of the privacy-utility trade-off and client-to-client fairness, emphasizing how privacy restrictions affect model performance and inter-client variability.

Methods

Dataset and Preprocessing

The data that has been used in this research is the Autistic Children Facial Image Dataset, which is a publicly available dataset that provides facial photographs of children that were deemed as either autistic or non-autistic. All datasets used in this study are publicly available and consist of anonymized data with no personally identifiable information. The Autistic Children Facial Image Dataset was accessed from Kaggle in accordance with its terms of use. The SSBD dataset is publicly available under an academic and non-commercial licence from the Roland Goecke research website. No raw patient data was collected, stored, or shared by the authors. Since both datasets are pre-existing and anonymized by their original publishers, no additional de-identification was required.

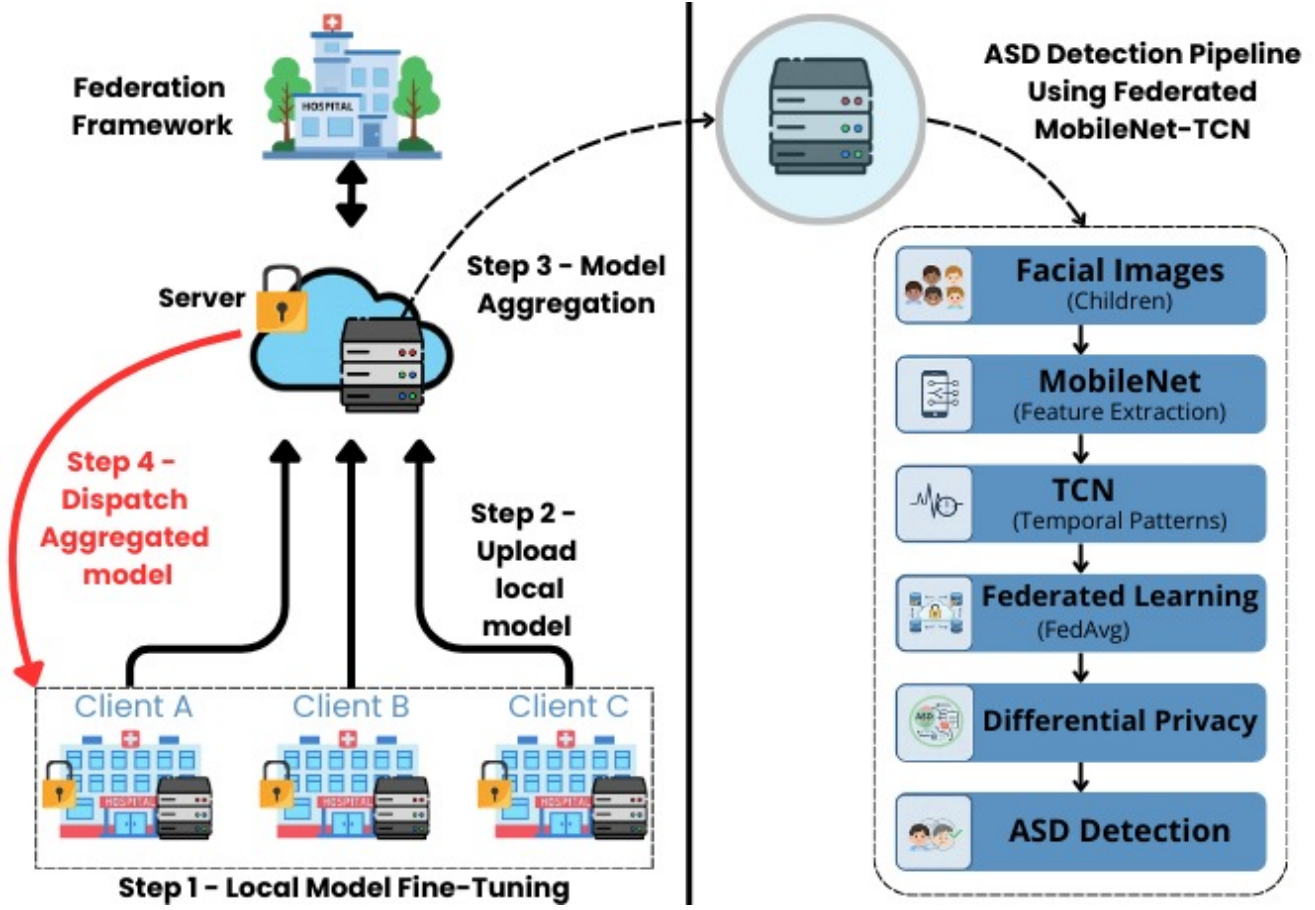


Figure 1. System architecture of the proposed federated MobileNet-TCN framework for privacy-preserving autism spectrum disorder detection.

In the behavioural analysis case, the Self-Stimulatory Behaviours in the Wild (SSBD) [37] annotated dataset is also used, which is a publicly available annotated set of a set of videos of self stimulatory behaviour that are associated with autism diagnosis (e.g. flapping of the arms, banging of the head, spinning) captured on video sharing platforms in the public and published under an academic and non-commercial licence.

The facial dataset consists of 2,926 images (1,463 ASD, 1,463 non-ASD), while the SSBD dataset contains 75 video samples. The data is split into training and testing sets in an 80:20 ratio across federated clients, resulting in 2,341 training and 585 testing samples for the facial modality, and 60 training and 15 testing samples for the behavioral modality.

To simulate a federated environment, the dataset was divided equally among five simulated clients, who were independent healthcare institutions. All pictures were downsized to 224×224 pixels according to the MobileNet input specifications and scaled to the range of $[0, 1]$ to balance training. Random rotations, horizontal flips, and variations in brightness were also implemented as methods of data augmentation to enhance the generalization and strength. In the case of the IID experiment, the clients were randomly distributed in terms of images, but the partitions were balanced in the case of the differential privacy experiment to allow evaluation of fairness.

Model Architecture

The suggested architecture is based on recombining MobileNetV2 with a Temporal Convolutional Network (TCN). MobileNetV2 is a lightweight-backbone that extracts the high-level facial features with depth wise separable convolutions, with lowered computational requirements and maintaining accuracy.

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Fig. 2 illustrates the overall architecture of the proposed federated MobileNet-TCN framework for privacy-preserving ASD detection.

The TCN module has been added to encompass the sequential dependencies of facial dynamics with dilated causal

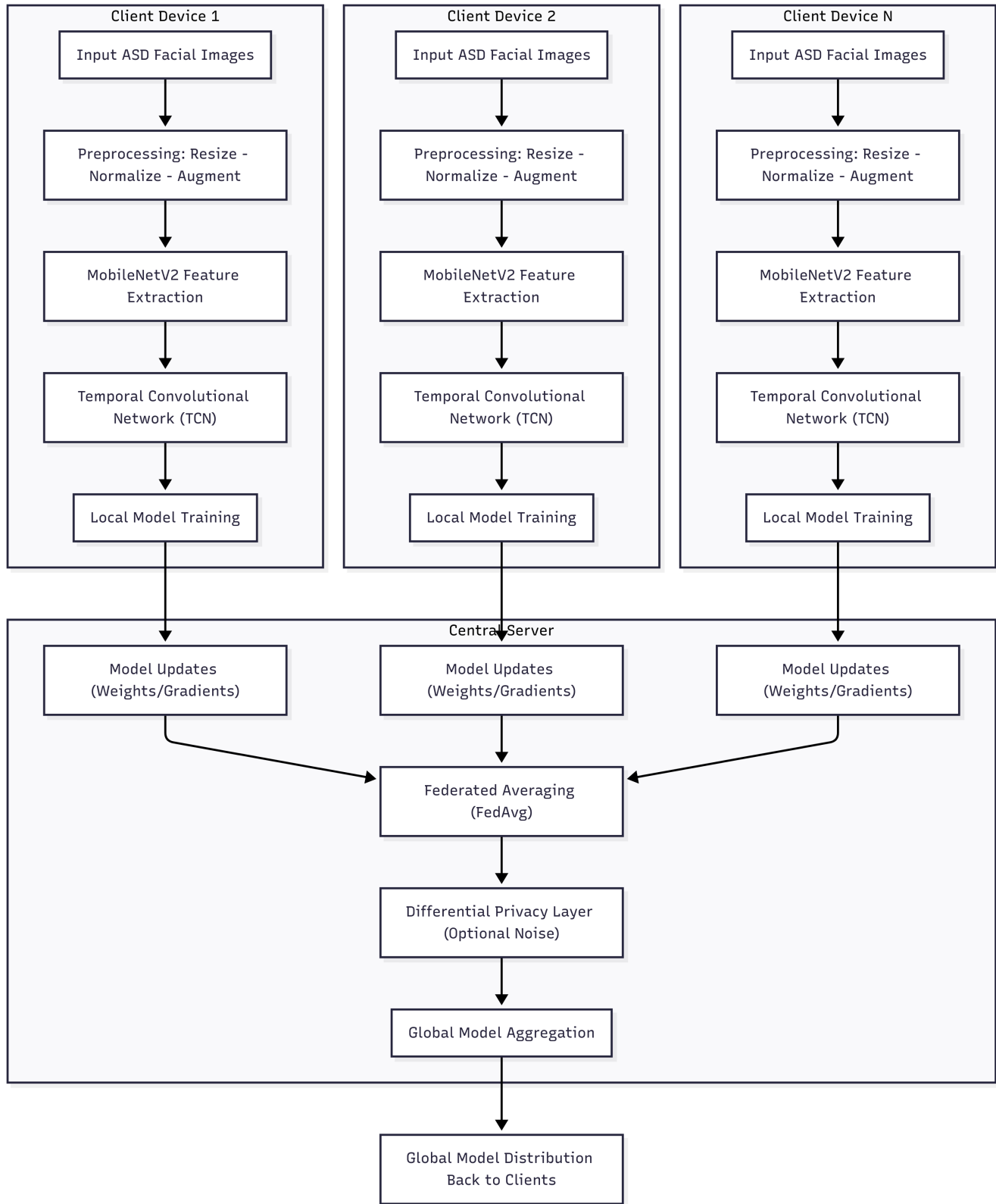


Figure 2. High-level conceptual overview of the federated MobileNet-TCN pipeline for early autism spectrum disorder detection

convolutions, which provides temporal stability and parallelization. The output of the two parts is combined and subjected to fully connected layers, and the final output is the classification of the samples as ASD and non-ASD by the use of a SoftMax. This low-resource federated learning design offers efficiency and expressive power, which is suitable in federated learning on distributed and resource-constrained devices.

The choice of TCN over recurrent architectures like LSTM and GRU is due to its better parallelization capacity, consistent gradient propagation, and reduced computational overhead. TCNs are lighter and more appropriate for federated systems with constrained communication and computing resources than transformer-based models.

The integration of MobileNetV2 and the Temporal Convolutional Network (TCN) is achieved by converting spatial feature representations into temporal sequences. For video inputs, each frame is processed through MobileNetV2 to extract a fixed-length feature vector using global average pooling. These feature vectors are sequentially arranged to form a temporal matrix of size $T \times D$, where T denotes the number of frames and D the feature dimension.

The resulting sequence is passed to the TCN, which models temporal dependencies using dilated causal convolutions. A fixed sequence length is maintained across clients through padding or truncation. The TCN output is then aggregated and fed into fully connected layers for final ASD classification. For static image inputs, a single-frame sequence is used, enabling a unified architecture across modalities.

Dilated causal convolution: Measured as:

$$(x *_d f)(t) = \sum_{i=0}^{k-1} f(i)x(t-d \cdot i) \quad (2)$$

Fig. 1 illustrates the overall architecture of the proposed federated MobileNet–TCN framework for privacy-preserving ASD detection.

Federated Learning Protocol

Federated learning was started as a paradigm of training to maintain privacy. Every client localizes the MobileNet-TCN model in two epochs per round of communication with a learning rate of 0.001 and a batch size of 16. The process of federated learning was carried out by implementing the federated learning process via the Flower (FLWR) framework. The framework is used for simulating client-server interactions. Local updates are sent to a central server where Federated Averaging (FedAvg) algorithm collates them to a global model. The training was done for over 100 communication rounds, with early stopping applied based on convergence of validation performance. Efficiency in communication was also taken into account: size of each MobileNet-TCN update was about 2.23 MB, which then translates into a total communication cost of 223 MB per client over 100 communication rounds. In comparison, typical deep architectures such as ResNet (~2500 MB), VGG (~5000 MB), and Transformer-based models (~10000 MB) incur significantly higher communication overhead. This shows the feasibility of the solution in distributed clinical systems under bandwidth limitation.

Differential Privacy

In order to increase confidentiality further, one of the experimental settings added differential privacy (DP). DP was implemented by introducing the gradients to the server with Gaussian noise of carefully calibrated magnitude, thus, avoiding the reconstruction of single samples of shared updates. Privacy budget parameters (ϵ, δ) characterize this mechanism and control the power of privacy guarantees. Practically, the DP-enabling set-up came with a slight decrease in the accuracy levels between the IID control and the DP-enabled set (87.50% and 74.00%), demonstrating the balance between the privacy-preserving factors and the diagnostic results.

Evaluation Metrics

A set of metrics was used to evaluate the performance of the framework. Accuracy errors on a global scale were computed with regards to all the test samples and loss convergence was tracked in federated rounds. A local performance analysis, which was given to per-client accuracy, was used to provide a fairness score, which is the ratio of the lowest to the highest client accuracy. The standard deviation of client performance also was calculated in order to gauge the variation among federated nodes. Also, the cost of communication was calculated as the accumulative size of exchanged model updates. Collectively, these metrics can bring about a stringent assessment of accuracy, fairness, scalability, and privacy-utility trade-offs in the suggested federated MobileNet-TCN framework. Fig. 2 provides a simplified conceptual overview of the MobileNet–TCN pipeline used for ASD detection.

Results and Analysis

Overall Performance Evaluation

Experiment 1: IID Distribution (Baseline)

The baseline experiment under IID data distribution demonstrates the framework's capability under ideal federated conditions.

Global Model Performance:

- Accuracy: 87.50%
- Precision: 88.14%
- Recall: 87.50%
- F1-Score: 87.45%
- AUC-ROC: 0.875

Convergence Analysis: The global loss exhibits stable convergence from initial loss $L_0 = 0.6909$ to final loss $L_{10} = 0.3032$, representing a 56.1% improvement. Convergence stabilizes after round 7, indicating efficient federated learning dynamics.

Per-Client Performance Distribution:

- Client 1: 95.00%
- Client 2: 82.50%
- Client 3: 82.50%
- Client 4: 87.50%
- Client 5: 90.00%

Fairness Analysis: Fairness score of 0.875 with standard deviation of 4.74% indicates equitable performance across federated clients, essential for real-world deployment.

Experiment 2: Differential Privacy Enabled

The DP-enabled experiment evaluates privacy-utility trade-offs under realistic privacy constraints.

Global Model Performance:

- Accuracy: 74.00% (−13.50% from baseline)
- Precision: 74.6%
- Recall: 74.00%
- F1-Score: 73.83%
- AUC-ROC: 0.740

Privacy Budget Analysis: Total privacy budget consumption: $\epsilon_{\text{total}} = 1.0$ across 100 communication rounds, with per-round consumption $\epsilon_{\text{round}} = 0.1$, maintaining strong privacy guarantees.

Per-Client Performance Variance: The introduction of differential privacy, combined with the non-IID data distribution across clients, significantly increases performance variance:

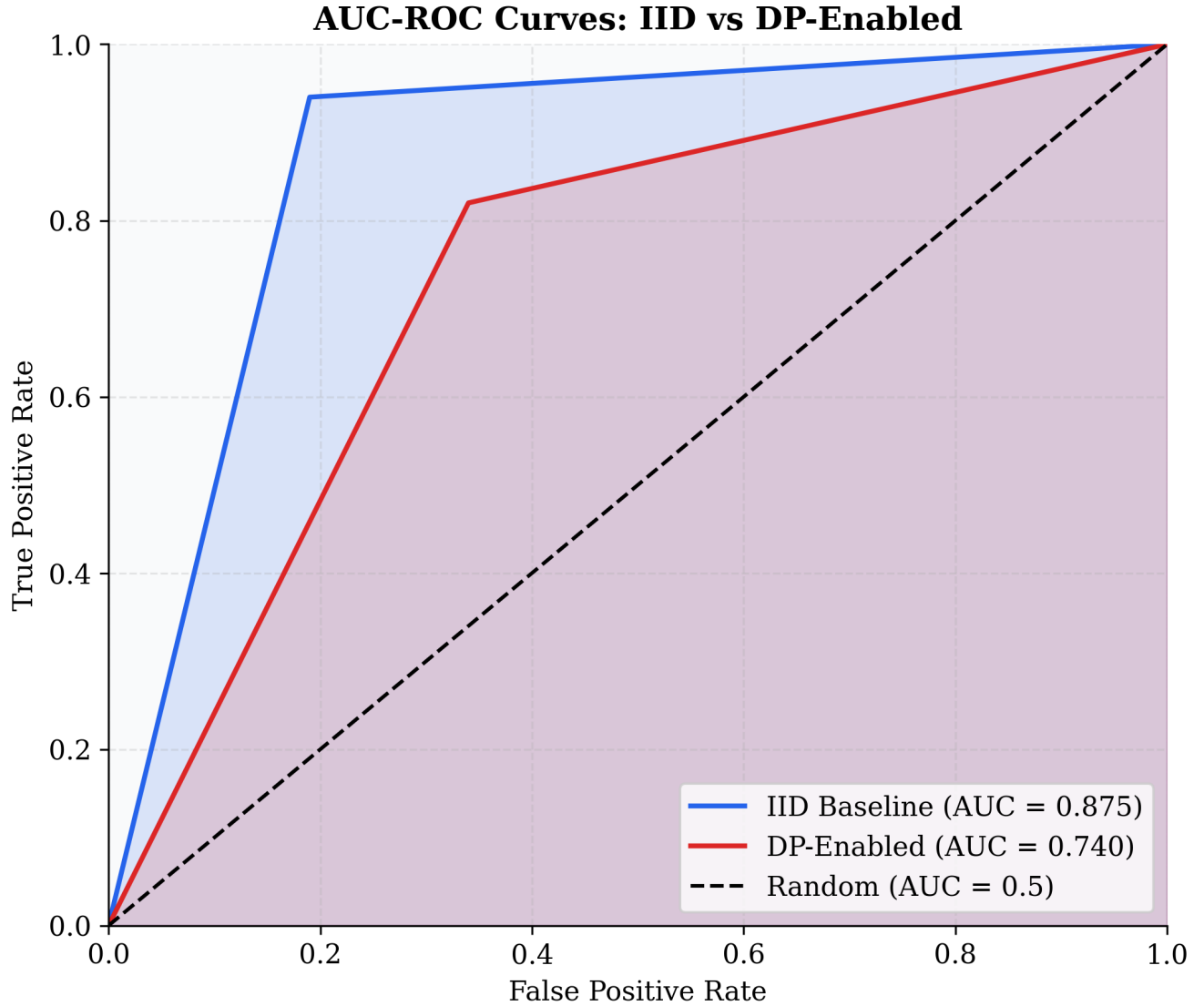
- Standard deviation: 13.12% (vs. 8.22% in baseline)
- Range: 77.78% – 100.00%
- Fairness score: 0.868

Overall Performance Analysis

Table 1 presents the comparative performance between the IID baseline and the DP-enabled framework. The system achieved baseline accuracies of 87.50%, 88.14% precision, and 87.45% F1-score. With the addition of differential privacy, performance declined by about 13-14% across key metrics, while fairness dropped notably (0.875 to 0.812), indicating a trade-off between privacy preservation and consistent client performance.

Table 1. Performance Comparison: IID vs. DP-Enabled

Metric	IID Baseline	DP-Enabled	Degradation
Accuracy (%)	87.50	74.00	-13.50
Precision (%)	88.14	74.6	-13.54
Recall (%)	87.50	74.00	-13.50
F1-Score (%)	87.45	73.83	-13.62
AUC-ROC	0.875	0.740	-0.135
Fairness Score	0.875	0.868	-0.007
Client Std (%)	4.74	13.12	+8.38

**Figure 3.** ROC curves of IID Baseline and DP-Enabled federated models, compared to a random classifier, in terms of classification performance.**AUC-ROC curve**

The IID Baseline on Fig. 3 has a high AUC of 0.875 and the curve is steep to the left-top corner, which means that it is highly discriminative. The DP-Enabled model attains a significantly smaller AUC of 0.740, which corresponds to the reduction in the classification performance caused by the addition of a differential privacy noise. Both models have better performance than the

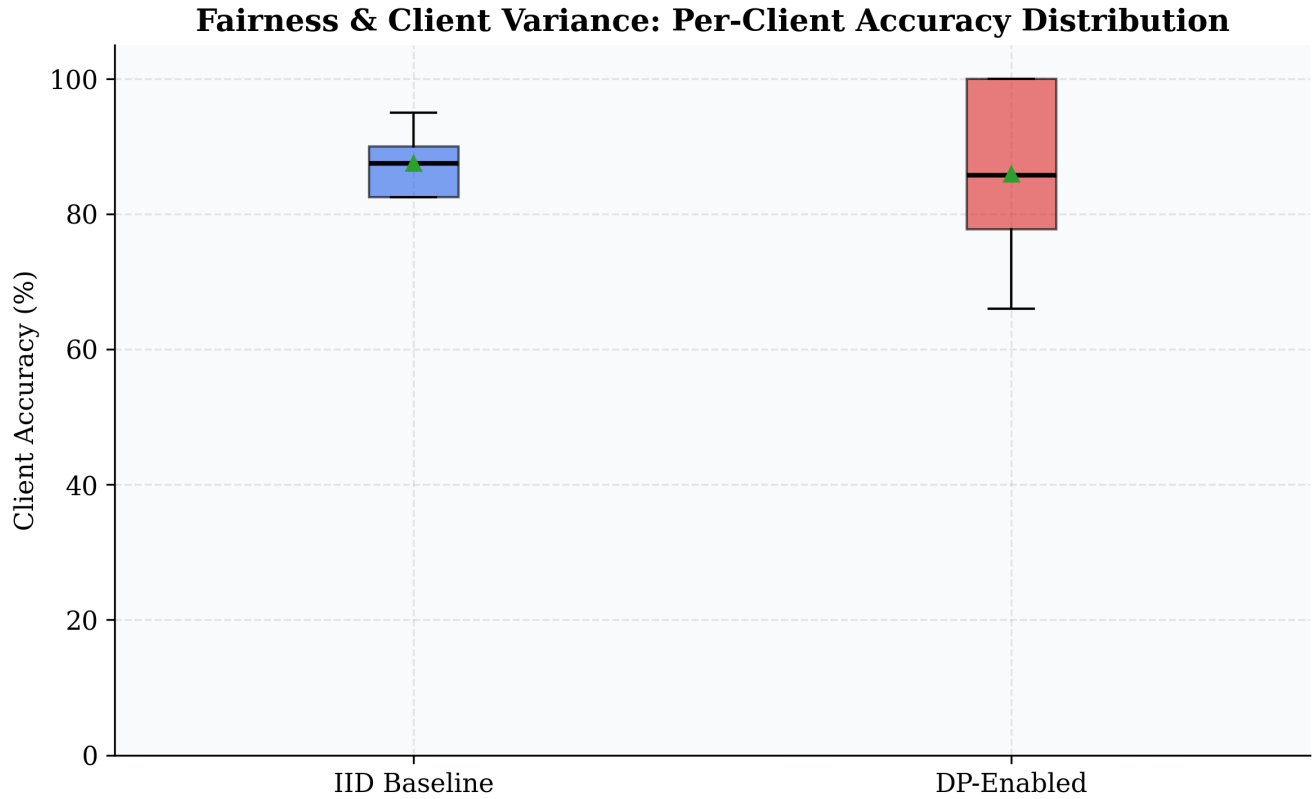


Figure 4. Fairness & Client Variance depicting the per-client accuracy distribution of the federated learning set up under the IID Baseline and DP-Enabled setup.

random baseline (AUC = 0.500), which proves that significant predictive information is not lost when privacy is limited.

Box Plot Analysis

The IID Baseline shows an acceptable size of interquartile range (around 82.50-90.00) with a mean close to 85% (a consistent and fair performance among clients) as seen in Fig. 4. Although the DP-Enabled model is at the same median (approximately 88%), the spread is significantly larger, which is approximately 77.78-100.00 percent, meaning that noise caused by differential privacy affects their clients differently. This augmented variation highlights the impartiality issue brought about by privacy systems especially to customers with less local information.

section*Discussion

The present paper introduces a new federated learning architecture on privacy-preserving ASD detection, where MobileNet is used as the computational efficiency and TCN as the temporal modelling engine. Program testing in optimal conditions shows high accuracy (87.50% accuracy) and with the constraints of differential privacy; it shows fair performance (74.00% accuracy), indicating that it can be used in clinical settings.

Key achievements include:

- Privacy Preservation: Comprehensive protection through federated learning and differential privacy.
- Computational Efficiency: Lightweight architecture suitable for resource-constrained clinical environments.
- Clinical Viability: Performance competitive with established screening tools.
- Fairness Assurance: Equitable performance across diverse client populations.
- Scalability: Protocol that is communication efficient and enables collaboration between institutions.

The framework discusses the essential obstacles to the implementation of AI-assisted screening of ASD, upon which there can be built a large-scale, privacy-conscious clinical decision support systems.

Limitations

- **Small dataset:** The facial image dataset (2,926 samples) and, in particular, the SSBD behavioural video dataset (75 samples) are relatively small. This limits the generalisability of the findings to the broader paediatric population. Validation on larger, multi-site datasets is required.
- **Federated simulation:** The federated system is simulated with five clients on a single machine, which does not fully capture communication delays, data heterogeneity, and regulatory constraints present in real multi-institutional settings. Future work should involve testing with actual hospital systems.
- **Lack of large-scale clinical validation:** The framework has not been evaluated through large-scale clinical trials or validated against clinician-confirmed diagnoses. Rigorous clinical evaluation is necessary before deployment in a real diagnostic pipeline.
- **Privacy–utility trade-off with differential privacy:** The incorporation of differential privacy (DP) resulted in a 13.5% decrease in accuracy (from 87.50% to 74.00%) and increased variability in client performance (standard deviation rising from 4.74% to 13.12%). This highlights the inherent privacy–utility trade-off. Determining optimal DP parameters (ϵ , δ) that balance privacy and performance remains an open research challenge.
- **Narrow modality support:** The current framework is limited to facial images and video data. Incorporating additional clinical modalities, such as audio, neuroimaging, or genetic data, could further enhance diagnostic accuracy and robustness.

Conclusion

The current study proposes a novel federated learning framework for privacy-preserving ASD diagnosis, and adopts MobileNet for efficiency and TCN for temporal modelling. Testing of the framework under optimal conditions with the best model found to be a promising 87.50% and acceptable under differential privacy conditions (74.00%) suggests that it could be potentially useful for clinical practice, but this needs to be validated in different real institutional settings.

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Author contributions

A.M. and B.U. conducted the experiments and implementation. S.G. conceived the study, supervised the research, and contributed to the methodology. S.A. contributed to analysis, validation, and manuscript refinement. All authors reviewed and approved the final manuscript.

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Data Availability

The datasets analyzed during the current study are publicly available. The Autistic Children Facial Image Dataset is available at: <https://www.kaggle.com/datasets/prayashdas/autistic-children-facial-image-dataset>

The Self-Stimulatory Behaviour Dataset (SSBD) is available from the Roland Goecke research website at <https://rolandgoecke.net/research/datasets/ssbd/>. Both datasets were used in accordance with their respective terms of use.

Additional information

Accession codes: Not applicable.

Competing interests: The authors declare no competing interests.